Ghazaleh Asghari PhD

Research Mathematician | Quantitative Research & Mathematical Modelling

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UK Global Talent visa (Exceptional Promise): no employer sponsorship required

PROFILE

Research mathematician with a PhD from the University of Reading, two published papers and a submitted preprint. Expertise in operator theory, spectral estimates and asymptotic analysis, supported by advanced training in probability and stochastic processes. Combines rigorous mathematical reasoning with Python-based algorithmic problem solving and project experience in Bayesian inference, statistical modelling and simulation-based trading decisions. Interested in quantitative research, trading, model development and validation, and mathematically demanding industry research.

QUANTITATIVE AND COMPUTATIONAL SKILLS

Probability & stochastic processes: PhD-level coursework in Markov chains, Poisson processes, Brownian motion, martingales, stopping times and Itô calculus; independent problem solving in conditional expectation and combinatorics.

Mathematical analysis: Operator and spectral theory; singular values and Schatten classes; Gaussian kernels and reproducing-kernel Hilbert spaces; stationary-phase and Laplace-type asymptotics.

Statistical modelling & decision methods: Project experience in Bayesian updating, online regression, Monte Carlo simulation, inventory-aware quote selection, bootstrap confidence intervals and held-out evaluation.

Programming & scientific tools: Python for algorithms and mathematical problem solving; working familiarity with NumPy, pandas, Matplotlib and Jupyter; regression and curve fitting from physics laboratory work; extensive LaTeX; exposure to C++.

QUANTITATIVE RESEARCH & TRADING PROJECTS

Dice Market-Making Agent | Everesteer Trading Challenge | October 2026

Python / NumPy

- Built the Bayesian inference and simulation core for a five-player, zero-sum dice market, estimating opponents' hidden information from quotes using online regression; integrated a neural likelihood.
- Selected quotes using 32 posterior-sampled states and exact order matching, accounting for adverse selection, stale-quote execution and quadratic inventory penalties under a 50 ms decision limit.
- Evaluated changes using recorded-state replay, separate selection and confirmation samples, blind time windows and match-level bootstrap confidence intervals. Finished 27th with Elo 1379 in the simulated-game challenge.

PROFESSIONAL EXPERIENCE

Scientific Reviewer, AI-Generated Mathematics and Physics | Mar 2026 - present

PI1 Labs, remote

- Assess research-level mathematics and theoretical physics generated by AI systems, checking correctness, assumptions and logical consistency.
- Identify gaps in technical arguments and evaluate whether conclusions are mathematically justified.

Doctoral Researcher | University of Reading, UK | 2022 - 2026

- Developed spectral and analytic methods for Toeplitz and Hankel operators and Berezin transforms; co-authored two published papers and one submitted preprint.
- Presented research at Oxford, King's College London, Lille, Toledo and IWOTA; refereed for three mathematical journals.

EDUCATION

PhD, Mathematics | University of Reading, UK | 2022 - 2026

Awarded May 2026. Advisor: Prof. Jani Virtanen.
Thesis: *Toeplitz, Hankel, and Berezin transforms on Fock-type spaces*. Spectral properties and Gaussian-type averaging on weighted spaces of holomorphic functions.

MSc, Mathematics | University of Zurich (UZH) and ETH Zurich | 2018 - 2021

Advisor: Prof. Will J. Merry. Thesis: *Fidelity and transversely intersecting Lagrangians*. Derived leading-order asymptotics for quantum-state fidelity using Bergman kernels and stationary phase.

MSc, Physics | Sharif University of Technology, Iran | 2015 - 2018

Advisor: Prof. Mohammad Mehdi Sheikh-Jabbari. Thesis: *Symplectic Symmetries and Their Associated Hilbert Spaces*. Symplectic geometry, geometric quantization and infinite-dimensional symmetry groups.

BSc, Physics, Minor in Mathematics | Sharif University of Technology | 2010 - 2015

Thesis: *Causal Sets Approach to Quantum Gravity*.

PUBLICATIONS

Fixed points of the Berezin transform on Fock-type spaces

With Ž. Čučković and S. Şahutoğlu. *Proc. Amer. Math. Soc.* 154 (2026), 3011-3026. arXiv:2508.13115

For weights $\exp(-|z|^m)$, polynomial fixed points are harmonic outside a possible countable exceptional set of $m > 0$.

Toeplitz operators on large vector-valued Fock spaces

With H. Arroussi and J. A. Virtanen. Preprint; submitted for publication (2025). arXiv:2504.15239

Derived boundedness, compactness and Schatten-class criteria for Toeplitz operators with positive operator-valued symbols on Hilbert-space-valued Fock spaces.

Schatten class Hankel operators on doubling Fock spaces and the Berger-Coburn phenomenon

With Z. Hu and J. A. Virtanen. *J. Math. Anal. Appl.* 540 (2024), 128596. Journal | arXiv:2312.06656

Characterised Schatten membership; proved the Hilbert-Schmidt Berger-Coburn property for bounded symbols and counterexamples in certain doubling Fock spaces for $0 < p \leq 1$.

In preparation: *Toeplitz operators on large Banach vector-valued Fock spaces* (sole author; final stages); *Sarason's conjecture on doubling Fock spaces* (with S. Miihkinen and J. A. Virtanen).

HONOURS AND AWARDS

EPSRC PhD Scholarship (2022-2026). Four-year award; sole EPSRC-funded PhD student in the department's 2022 intake.

SwissMAP Equal Opportunities Excellence Scholarship (2018-2020).

Excellence Master Fellowship, University of Geneva (2018). Awarded annually to approximately 16 Master's students across all disciplines.

SELECTED TALKS

Workshop on Operator Theory and Operator Algebras, University of Reading (invited speaker; Sep 2026); Université de Lille, Functional Analysis Seminar (Apr 2026); University of Oxford, Functional Analysis Seminar (Nov 2025); IWOTA, University of Twente (Jul 2025); King's College London, Analysis Seminar (Jun 2024); University of Toledo, Complex Analysis & Operator Theory Seminar (Apr 2024).

TEACHING AND ACADEMIC SERVICE

Journal referee (2023-present): *J. Math. Anal. Appl.*, *Complex Var. Elliptic Equ.* and *Bull. Sci. Math.*
Co-organiser, SIAM-IMA Reading Student Chapter seminar series (2024-2026).
Substitute teacher, Calculus in One Variable, KTH Royal Institute of Technology (2023).
Teaching assistant, University of Reading: real and complex analysis, analytic number theory and foundations (2022-2024).
Seminar design and teaching, *Topology from the Differentiable Viewpoint*, advanced undergraduate and Master's students, University of Reading (2022).